



Guidelines

Sleep Optimisation

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	All	All

Introduction

Sleep is important for recovery from illness as it aids in the body's repair process and the building of immunity. It also assists with managing pain and reducing the stress of being in hospital ¹. However, the hospital environment can make it challenging for patients to have good quality sleep.

SCGOPHCG is committed to providing an environment that optimises sleep in the hospital setting to enhance patient experience and facilitate recovery.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

This guideline is applicable to non-critically ill hospitalised patients and aims to:

- Support clinicians in optimising patients' sleep through environmental and non-pharmacological strategies.
- Reduce harm from use of medicines for night sedation.

Sleep optimisation strategies

There are many factors that may disturb sleep of patients in hospital, such as the hospital/ward environment, patients' physical illness, emotional state, and staff care interventions and activities ². Sleep optimising strategies are focused on how to modify these factors so that sleep is less disturbed for patients.

1.0 Non-pharmacological Management

Non-pharmacological therapy is recommended as the first-line management option for sleep disturbance and insomnia in hospitalised patients.

1.1 Environmental Strategies

Environmental factors play a significant role in influencing the quality and quantity of patients' sleep. Noise and lighting are modifiable causes of some sleep disruptions in hospitals and when reduced can lead to more sleep.

Reduce light and noise wherever possible:

- Lights should be turned down or off at night time. If shades are used, they should be closed and directed lights or torches should be used where necessary.
- Minimise disruptive noises like alarms and pagers, and reduce device and appliance movement.
- Promote physical comfort and sleep aids (e.g. extra blankets, earplugs and eye masks.)
- Use silent mode on mobile phones and pagers at night.
- Ensure staff conversations are away from patient bed spaces during the night and be mindful of the volume level.
- In shared rooms, during quiet/nighttime, restrict the viewing of television and other devices and encourage the use of headphones where possible.
- Consider allocating patients who may disrupt others' sleep to single rooms.
- Display posters highlighting the importance of quietness at night in the ward.

1.2 Care Intervention/Activities

Nighttime care interventions are recognised as a major contributor to sleep disruption in hospitals. Below strategies are recommended to cluster care activities if possible, and only provide necessary care interventions at nighttime in consultation with the treating team.

- Patient care interventions to occur during awake time whenever possible.
- Non-acute patient observation may be decreased by registrar/consultant following patient assessment; applicable to patients awaiting permanent residential care, permanent aged care residents and end of life / palliative patients. Changes in observation frequency must be documented in the patient's medical record and be consistent with the patients' clinical situation and Goals of Patient Care (GoPC).
- Avoid early morning awakening (before 6am), as this is when restorative sleep occurs.
- Patients are not administered prescribed night sedation medications, if asleep.
- For TDS or QID medications, consider the timing written in the drug chart and whether adjustments can be made (i.e. QID paracetamol dosing could be 0600/1200/1800/2200 hours)

1.3 Other

- Daylight and physical activity are essential to promote a good night time sleep:
 - encourage patients to do light physical activities (e.g. exercise in the bed or chair) when appropriate.
 - The physiotherapists in the hospital will assist with rehabilitation programs.
 - Where possible, staff should assist with enabling patients to see some sunlight.
- Patient Education:
 - Reduce stimuli at night e.g. television, phones and computers.
 - Provide information about the value of sleep health.
 - Facilitate shared decision making.

2.0 Pharmacological management

Initiation of medicines for management of sleep disturbance or insomnia in hospitalised non-critically ill patients is not recommended and should only occur after comprehensive assessment, optimisation of non-pharmacological management, discussion of the limited effectiveness and significant harms of these medicines.

Effectiveness and safety should be evaluated prior to prescribing medications to improve sleep in patients. Below are key principles for medications and sleep, and should be adhered to wherever possible.

2.1 Principles for medications and sleeping

- Consider other causes of poor sleep in the patient and address those (pain, anxiety, environmental considerations, CNS-stimulating medications such as steroids)
- Benzodiazepines and Z-drugs (zolpidem, zopiclone) are low-value, high-risk medications for sleep disturbance or insomnia in hospital, often causing preventable harm.
- The use of other medicines for night sedation (e.g. melatonin, antipsychotics, antihistamines, antidepressants, suvorexant) is not recommended, as the evidence does not favour benefit over harm in this setting.
- If medicines for night sedation are selected as part of a management plan for sleep disturbance or insomnia in hospital, despite the risk of harm or lack of effectiveness, the following actions are recommended:
 - Clinicians should discuss sleep hygiene principles at every opportunity.
 - Communicate the limited effect and significant harm of medicines for night sedation to patients/carers, obtain informed consent and prescribe for an agreed period.
 - Check for aberrant drug-related behaviours and history of substance use disorder.
 - Beware of prescribing medicines for night sedation with other central nervous system depressants, such as opioids and gabapentinoids.
 - Be aware of the factors that might make a patient more sensitive to the effects of the medication (age, eGFR, liver function, other CNS acting agents).
 - Review use and impact of medicine for night sedation daily.
 - In all patients who have taken medicines for night sedation prior to hospital admission, refer for a medication review and discuss gradually deprescribing night sedation medicines at every opportunity. If deprescribing is commenced in hospital, wean slowly and monitor for withdrawal symptoms. Also communicate this to the patient's GP.
 - In previously treatment-naïve patients, do not continue medicines for night sedation after discharge.

2.2 Short-term Pharmacological options for promoting sleep in patients

- Temazepam 5-10mg PRN is recommended as first line treatment for insomnia. For adults > 70 years 5mg would be an appropriate starting dose.
- Prescription is for 1-3 day trial period only (unless used prior to hospitalisation). If the medication is to be continued after the trial period:
 - The success and ongoing plan must be documented in the medical record.
 - The prescription should not exceed 2 weeks and should not be on discharge from hospital unless an explicit reason is documented (e.g. oncology patient on high-dose steroids).

Night sedation medicines such as benzodiazepines and Z-drugs (zolpidem, zopiclone) improve sleep quality in 1 in 13 patients while adverse effects occur in 1 in 6 patients.³

Table1: Risk factors for adverse outcomes with benzodiazepines and Z-drugs use. If these risk factors are present consider reducing dose or trial of melatonin as an alternative.

Use of medication and other drugs	Presence of morbidities	Geriatric associated factors
At initiation; previous high dose or prolonged use (>4 weeks) of benzodiazepines/Z-drugs	Respiratory disease of pulmonary insufficiency, hepatic and renal impairment.	Age >65 years
Concomitant use of interacting medicines (regular and as needed (PRN))	Mental health condition (e.g. depression, psychosis or schizophrenia). There is potential for the condition to worsen (including risk of suicide). N.B. combining zolpidem with drugs used to treat these conditions may also increase the likelihood of dangerous behaviour e.g. sleepwalking.	Exciting cognitive impairment
Concomitant use of alcohol or other illicit drugs		Pre-existing falls risks

References: AMH Aged Care Companion⁴ & RACGP Prescribing drugs of dependence in general practice Part A.⁵

Zopiclone and zolpidem

The Statewide Formulary restricts these agents to situations where the patient has not responded to, or are intolerant of benzodiazepines, and must be directed by a psychiatrist, for use less than 14 days. Where these conditions are not met, an individual patient approval (IPA) from the Drug and Therapeutics Committee is required for initiation of use in hospital.

Melatonin

Melatonin is TGA approved for short term monotherapy treatment of insomnia (characterised by poor sleep quality) in adults over 55 years of age. It is not listed on the Pharmaceutical Benefit Scheme (PBS) and requires an individual patient approval (IPA) for initiation of use in hospital for this indication.

The pivotal trials supporting the registration of modified release (MR) melatonin in Australia for use in primary insomnia showed only modest improvements in sleep and a substantial proportion of subjects (74%) experienced no improvement. The MR melatonin formulation is designed to mimic endogenous melatonin concentrations; however, its delayed onset of action makes it less likely to be beneficial for those having difficulty getting to sleep.

The dose for MR melatonin is 2mg nocte. There is also an immediate release version (3mg) that can be doubled to 6mg as needed.

While melatonin participates in the regulation of the sleep-wake cycle, its role as a therapeutic drug in promoting sleep is controversial with contradictory or negative results seen in numerous studies.⁶

2.3 Deprescribing

While there are advantages to considering deprescribing in hospital, the acute stresses of hospitalisation and the risk of rebound insomnia during hospitalisation may mean that initiation of a deprescribing plan after hospital discharge may be more appropriate in some

patients. Clear communication with the clinician taking over care after discharge is required, including what the recommendation / plan is on discharge.

Abrupt cessation of benzodiazepines can cause withdrawal symptoms (e.g. anxiety, irritability, insomnia, nightmares, changes to memory and concentration) which are usually mild, highly variable and can last up to 6-8 weeks. Withdrawal symptoms are more likely and more serious (e.g. convulsions) when benzodiazepines are used at high doses, involve high potency benzodiazepines (e.g. alprazolam) or have been used for a prolonged period of time.

[Refer to Deprescribing Guide for Benzodiazepines and Z Drugs for guidance.](#)

2.4 Patient counselling

Discussions with the patient regarding the poor effectiveness and significant potential harms of night sedation medicines and information about deprescribing should be undertaken by the treating team.

Provide consumers with information leaflets (e.g. Sleep and sleep hygiene strategies and deprescribing of night sedation medicines.)

Consider referring patients with chronic insomnia for Cognitive Behavioral Therapy for Insomnia (CBT-I) or brief behavioural therapy.

Related Documents

[SCGOPHCG Sleep poster](#)

[Sleeping well in hospital patient flyer](#)

[Sleep medication patient flyer](#)

[NSW Government - Deprescribing guide for Benzodiazepines and Z drugs](#)

References

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